

24ITT203	COMPUTER ORGANIZATION AND ARCHITECTURE (Common to CS, IT)	L	T	P	J	C
		3	0	0	0	3
		SDG		4,9		
PC						

Pre-requisite courses	Nil	Data Book / Codes / Standards (If any)	Nil
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Course Objectives:	
The purpose of taking this course is to:	
1	Introduce the basic structure and functional components of a computer system, including its evolution, Von Neumann architecture, and performance parameters.
2	Explain the internal organization of a basic computer, focusing on instruction formats, addressing modes, and the distinction between CISC and RISC architectures.
3	Explore arithmetic and logic unit (ALU) operations including integer and floating-point computations.
4	Examine memory hierarchy, cache, and I/O system designs and their impact on performance.
5	Explore modern advancements in computer architecture, including multicore processors, parallel architectures, and multiprocessor systems with interconnection structures and synchronization techniques.

Course Outcomes:		
After successful completion of this course, the students shall be able to		Bloom's Taxonomy Level (BTL)
CO 1	Identify and explain the structure, functional units, performance characteristics of computer systems	U
CO 2	Apply arithmetic algorithms to perform for addition, subtraction, multiplication, and division with correctness and efficiency in problem-solving tasks.	Ap
CO 3	Apply the concepts of instruction execution, multiple bus organization, and control unit design to implement control logic using hardwired and microprogrammed approaches.	Ap
CO 4	Explain the operations of I/O subsystems and memory architectures, and describe the roles of programmed I/O, interrupts, DMA, RAM, cache, and memory hierarchy in data transfer and system performance.	U
CO 5	Utilize the principles of pipelining, hazard management, and parallel processing architectures such as superscalar, multicore, and GPU acceleration to improve performance in real-time computing systems.	Ap

	Program Outcomes (PO) (Strong-3, Medium – 2, Weak-1)											Program Specific Outcomes (PSO)	
	1	2	3	4	5	6	7	8	9	10	11		
Course Outcomes (CO)	Engineering Knowledge	Problem Analysis	Design/Development of Solutions	Conduct Investigations of Complex Problems	Engineering Tool Usage	The Engineer and The World	Ethics	Individual and Collaborative Team work	Communication	Project Management and Finance	Life-Long Learning	PSO-1	PSO-2
1	2	2											
2	2	2											
3	3		2									2	
4	3	2										2	
5	2	2										1	

Course Content	
FUNDAMENTALS OF COMPUTER ORGANIZATION Organization and Architecture- Structure and Function- Functional Units-Basic Operations and Concepts-Bus Structure-Performance. Memory Locations and Addresses - Memory Operations - Instruction and Instruction Sequencing - Addressing Modes - Basic I/O Operations. Case Study: Evolution of Intel, ARM Architectures/Processor Families	7 Hours
COMPUTER ARITHMETIC Addition and Subtraction of Signed Numbers-Design of Fast Adders-Multiplication of Positive Numbers-Signed Operand Multiplication-Fast Multiplication-Integer Division-Floating Point Numbers and Operation.	11 Hours
BASIC PROCESSING UNIT Fundamental Concepts - Execution of a Complete Instruction - Multiple Bus Organization - Hardwired Control – Microprogrammed Control – Microinstructions-Microprogram Sequencing-Wide Branch Addressing	9 Hours
I/O AND MEMORY The I/O Subsystem, -Programmed I/O- I/O Interrupts -Direct Memory Access (DMA) - The Components of Memory System – RAM Structure: The Logic Designer’s Perspective – Memory Boards and Modules-Memory Hierarchy – The Cache- The Memory Subsystem.	9 Hours
PIPELING AND PARALLEL PROCESSING Basic Concepts - Data Hazards - Instruction Hazards – Influence on instruction sets - Data path and control considerations - Superscalar operation. Multiple Processor Organization-Types, Parallel Organization. Multicore Computers Introduction – Organization- Software and Hardware performance issues. Case Study: GPU Acceleration in Real-Time System.	9 Hours

Theory Hours:	45	Tutorial Hours:	0	Practical Hours:	0	Project Hours:	0	Total Hours:	45
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Learning Resources	
Textbooks	
1. Carl Hamacher, Zvonko Vranesic and Safwat Zaky, Computer Organization, 5th Edition, McGraw-Hill, (2014). 2. Vincent P Heuring, Harry F Jordan, TG Venkatesh., Computer Systems Design and Architecture, 2nd Edition, Pearson, (2014). 3. William Stallings, Computer Organization and Architecture - Designing for Performance, 9th Edition, Prentice Hall, (2012).	
Reference books/ Web Links	
1. M. Moris Mano, Computer System Architecture, Revised 3 rd Edition, Pearson, (2017). 2. John P.Hayes, Computer Architecture and Organization, 3 rd Edition, McGraw Hill, (2002). 3. David A.Patterson and John L.Hennessy, Computer Organization and Design: The hardware / software interface, 5th Edition, Morgan Kaufmann, (2014). 4. Andrew S. Tanenbaum, Structured Computer Organization, 6 th Edition Pearson Education, (2016)	
Online Resources (Weblinks)	
1. https://profile.iiita.ac.in/bibhas.ghoshal/teaching_coa_2021.html 2. https://www.cse.iitd.ac.in/~srsarangi/archbooksoft.html 3. https://onlinecourses.nptel.ac.in/noc22_cs88/preview 4. https://www.coursera.org/learn/comparch 5. https://www.udemy.com/course/computer-organization-and-architecture-j/?srsltid=AfmBOoo3L-F9CWPP3HioEoAzx57cX2iGGEU3Bi0UOLTD0aLRh_BAREfE&couponCode=ST21MT30625G1	

Assessment (Theory course)
SA-I, SA-II, Activity and Learning Task(s), MCQ, End Semester Examination (ESE)

Course Curated by			
Expert(s) from Industry	Expert(s) from Higher Education Institution		Internal Expert
-	-		Mr. M. Sathish, Department of Computer Science and Engineering
Recommended by BoS on	09.05.2025		
Academic Council Approval	No: 28	Date	26.06.2025